

Mathematics A

Unit 1 • Chapter OL-T17 • Expertise Q16+ Classified

Cross-session • chapter-organised candidate

3 classified items • 11 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

Every question is followed by its worked answer/reference

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Contents

Unit 1 • Expertise • Unit 1 • Chapter OL-T17 • Expertise Q16+ • 3 items • 11 marks

Chapter	Items	Marks	Starts
01. OL-T17 Algebraic Roots & Indices	3	11	4

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Algebraic Roots & Indices

Unit 1 • Expertise • 11 marks • 3 item(s)

June 2024 | 4MA1-2H Q23 | 3 marks

Complex indexed monomial or fractional outer power

June 2022 | 4MA1-2H Q24 | 3 marks

Express one exponent variable in terms of others

January 2023 | 2H Q26 | 5 marks

Zero and negative indices

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

4MA1-2H Q23

3 marks

23 Simplify $\frac{30 \times 25^{2x+7}}{\sqrt{180} \times (\sqrt{5})^{4x+9}}$

Give your answer in the form 5^w where w is an expression in terms of x
Show each stage of your working clearly.

WORKING SPACE

4MA1-2H Q23

3 marks

23 Simplify $\frac{30 \times 25^{2x+7}}{\sqrt{180} \times (\sqrt{5})^{4x+9}}$

Give your answer in the form 5^w where w is an expression in terms of x
Show each stage of your working clearly.

Surds and index laws

3 marks

Rewrite in base 5

$$30 = 6 \cdot 5, 25^{2x+7} = 5^{4x+14}, \sqrt{180} = 6\sqrt{5}.$$

Combine

$$\frac{30 \cdot 25^{2x+7}}{\sqrt{180}(\sqrt{5})^{4x+9}} = 5^{2x+10}.$$

Final answer

$$5^{2x+10}$$

4MA1-2H Q24 Q24

3 marks

24

$$\frac{18 \times (\sqrt{27})^{4n+6}}{6 \times 9^{2n+8}} = 3^x$$

Express x in terms of n

Show your working clearly and simplify your expression.

 $x = \dots\dots\dots$

WORKING SPACE

4MA1-2H Q24 Q24

3 marks

24

$$\frac{18 \times (\sqrt{27})^{4n+6}}{6 \times 9^{2n+8}} = 3^x$$

Express x in terms of n

Show your working clearly and simplify your expression.

 $x = \dots\dots\dots$

Chapter: Algebraic Roots & Indices

3 marks

Q24 – Chapter: Algebraic Roots & Indices

Family: Solve an index relation by a common base • Variant: Express one exponent variable in terms of others

Method

Use $18 = 2 \times 3^2$, $\sqrt{27} = 3^{(3/2)}$, $6 = 2 \times 3$, and $9 = 3^2$.

Why: Rewrite every factor using base 3 where possible.

Method

The left side becomes $[2 \times 3^2 \times 3^{(6n+9)}] / [2 \times 3 \times 3^{(4n+16)}]$.

Why: Multiply the nested exponents and collect the fixed factors.

Method

This simplifies to $3^{(6n+11-4n-17)} = 3^{(2n-6)}$, so $x = 2n - 6$.

Why: Subtract denominator exponents and equate equal bases.

Final answer

x: $x = 2n - 6$

26 Find the values of n such that

$$\frac{10^{4n} \times 2^{3(n^2-5n)} \times 5^{2(1-2n)}}{20^2} = 1$$

Show clear algebraic working.

(Total for Question 26 is 5 marks)

Answer reference | January 2023 | Question 26

Family: Interpret zero, negative and fractional indices • Variant: Zero and negative indices

Q	working	Answer	Mark	Notes
26	For one of: $10^{4n} = (5 \times 2)^{4n}$ or $5^{4n} \times 2^{4n}$ or oe $20^2 = (5 \times 2^2)^2$ or $5^2 \times 2^4$ or $5^2 \times 2^2 \times 2^2$ or cancelling 5^2 on numerator with 20^2 to get 16 or 2^4		5	M1 for writing 10^{4n} correctly as a product of 5 and 2 to the power $4n$ oe or writing 20^2 correctly as a product of 5 and 2^2 to the power 2 oe or cancelling 5^2 on numerator with 20^2 to get 16 or 2^4
	for getting numerator to the stage (this scores M1M1) $2^{4n} \times 2^{3n^2-15n} \times 5^2$ oe eg $2^{3n^2-11n} \times 5^{-4n+4n+2}$ oe or for two of $10^{4n} = (5 \times 2)^{4n}$ or $5^{4n} \times 2^{4n}$ or oe $20^2 = (5 \times 2^2)^2$ or $5^2 \times 2^4$ or $5^2 \times 2^2 \times 2^2$ or cancelling 5^2 on numerator with 20^2 to get 16 or 2^4 or getting the numerator to the stage			M1
	$\frac{2^{4n} \times 2^{3n^2} \times 2^{-15n}}{2^4} [= 1 \text{ or } 2^0]$ or $2^{4n} \times 2^{3n^2} \times 2^{-15n} = 2^4$ oe			M1 For writing the equation in powers of 2 only
	$3n^2 - 11n - 4 [= 0]$			A1 Correct quadratic equation dep on M1
	working required	$-\frac{1}{3}, 4$		A1 Both answers required dep on M1
				Total 5 marks

Worked solution

January 2023 | Question 26 | 5 marks

Family: Interpret zero, negative and fractional indices • Variant: Zero and negative indices

Official final mark-scheme pages are embedded immediately after this question.